

```
!pip install transformers datasets torch scikit-learn pandas
```

```
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Requirement already satisfied: datasets in /usr/local/lib/python3.12/dist-packages (4.0.0)
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```

```
import torch
from transformers import AutoTokenizer, AutoModelForSequenceClassification, Trainer, TrainingArgu
from datasets import load_dataset
```

```
from sklearn.metrics import accuracy_score, precision_recall_fscore_support
import numpy as np
```

```
dataset = load_dataset("imdb")

print(dataset)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens)
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub.
README.md:      7.81k/? [00:00<00:00, 188kB/s]

plain_text/train-00000-of-100000.parquet: 100% 21.0M/21.0M [00:00<00:00, 105MB/s]

plain_text/test-00000-of-100000.parquet: 100% 20.5M/20.5M [00:00<00:00, 102MB/s]

plain_text/unsupervised-00000-of-100000.p(...): 100% 42.0M/42.0M [00:00<00:00, 109MB/s]

Generating train split: 100% 25000/25000 [00:00<00:00, 83628.70 examples/s]
Generating test split: 100% 25000/25000 [00:00<00:00, 108054.28 examples/s]
Generating unsupervised split: 100% 50000/50000 [00:00<00:00, 124394.53 examples/s]

DatasetDict({
  train: Dataset({
    features: ['text', 'label'],
    num_rows: 25000
  })
  test: Dataset({
    features: ['text', 'label'],
    num_rows: 25000
  })
  unsupervised: Dataset({
    features: ['text', 'label'],
    num_rows: 50000
  })
})
```

```
print(dataset['train'][0])
```

```
{'text': 'I rented I AM CURIOUS-YELLOW from my video store because of all the controversy that sur
```

```
tokenizer = AutoTokenizer.from_pretrained("distilbert-base-uncased")
```

```
config.json: 100% 483/483 [00:00<00:00, 45.5kB/s]

tokenizer_config.json: 100% 48.0/48.0 [00:00<00:00, 3.63kB/s]

vocab.txt: 232k/? [00:00<00:00, 6.94MB/s]

tokenizer.json: 466k/? [00:00<00:00, 14.4MB/s]
```

```
def tokenize_function(example):
    return tokenizer(example["text"], padding="max_length", truncation=True)
```

Map: 100%	25000/25000 [00:30<00:00, 824.77 examples/s]
Map: 100%	25000/25000 [00:37<00:00, 808.85 examples/s]
Map: 100%	50000/50000 [01:00<00:00, 889.97 examples/s]

```
print(tokenized_datasets["train"][0])
```

[illegible]

[illegible]

```
small_train_dataset = tokenized_datasets["train"].shuffle(seed=42).select(range(2000))
small_test_dataset = tokenized_datasets["test"].shuffle(seed=42).select(range(1000))
```

```
model = AutoModelForSequenceClassification.from_pretrained("distilbert-base-uncased", num_labels=
```

model.safetensors: 100%

268M/268M [00:01<00:00, 264MB/s]

Loading weights: 100% 100/100 [00:00<00:00, 372.77it/s, Materializing param=distilbert.transformer.layer.5.sa_layer_norm.w

DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased

Key	Status
vocab_transform.weight	UNEXPECTED
vocab_projector.bias	UNEXPECTED
vocab_transform.bias	UNEXPECTED
vocab_layer_norm.weight	UNEXPECTED
vocab_layer_norm.bias	UNEXPECTED
classifier.bias	MISSING
classifier.weight	MISSING
pre_classifier.weight	MISSING
pre_classifier.bias	MISSING

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expected
- MISSING :those params were newly initialized because missing from the checkpoint. Consider training

```
def compute_metrics(pred):
    labels = pred.label_ids
    preds = np.argmax(pred.predictions, axis=1)

    precision, recall, f1, _ = precision_recall_fscore_support(labels, preds, average='binary')
    acc = accuracy_score(labels, preds)

    return {
        'accuracy': acc,
        'f1': f1,
        'precision': precision,
        'recall': recall
    }
```

```
training_args = TrainingArguments(
    output_dir="./results",
    eval_strategy="epoch",
    learning_rate=2e-5,
    per_device_train_batch_size=8,
    per_device_eval_batch_size=8,
    num_train_epochs=2,
    weight_decay=0.01,
)
```

```
trainer = Trainer(
    model=model,
```

```

args=training_args,
train_dataset=small_train_dataset,
eval_dataset=small_test_dataset,
compute_metrics=compute_metrics
)

```

```

text = "This movie was absolutely amazing!"

inputs = tokenizer(text, return_tensors="pt", truncation=True, padding=True)

outputs = model(**inputs)

pred = torch.argmax(outputs.logits)

if pred == 1:
    print("Positive Sentiment")
else:
    print("Negative Sentiment")

```

Positive Sentiment

```

import pandas as pd

# Evaluate the model
results = trainer.evaluate()

# Create table
data = {
    "Metric": ["Accuracy", "Precision", "Recall", "F1 Score"],
    "Score": [
        results["eval_accuracy"],
        results["eval_precision"],
        results["eval_recall"],
        results["eval_f1"]
    ]
}

result_table = pd.DataFrame(data)

print(result_table)

```

```

/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memory' is deprecated. Please use 'pin_memory_device' instead.
  super().__init__(loader)
[ 84/125 09:41 < 04:47, 0.14 it/s]

```

